

This is a copy of a conversation between ChatGPT & Anonymous.

Report content

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Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.

I've received your dataset: `sicor_municipal_contracts.csv`. I'll first inspect its structure and key characteristics before proceeding with any specific analysis. Let me load and summarize the data.

The dataset consists of **231,636 records** and **19 columns**, containing information on municipal contracts. Here are the key characteristics:

- **Identifiers:** `StateCode`, `MunicipalityCode`, `ProgramCode`, `SubProgramCode`
- **Time-related data:** `MonthIssued`
- **Financial Details:**
 - `FundingValue`, `InvestmentValue`, `CommercializationValue`, `IndustrializationValue`
 - `ContractTotalValue` : Overall value of the contract
- **Quantities:** `FundingQuantity`, `InvestmentQuantity`, `CommercializationQuantity`, `IndustrializationQuantity`
- **Land Allocation:** `AllocatedAreaInvestment`, `AllocatedAreaFunding`
- **Financial Institution:** `FinancialInstitutionCNPJ`
- **Economic Activity:** `EconomicActivity`

Can a model predict the most used program code for each month of next year in the state with the highest variation in its most frequent program, based on historical contract granting patterns? Define three approaches to address this issue and select the best one.

This is a forecasting problem where we aim to predict the most used **ProgramCode** for each month of next year in the state that has the **highest variation** in its most frequent program.

Step 1: Identifying the State with the Highest Variation in its Most Frequent Program

1. Determine the **most frequent** `ProgramCode` for each state over time.
2. Measure the **variation** (e.g., standard deviation, coefficient of variation) in this dominant program's frequency.
3. Identify the **state with the highest variation**.

Step 2: Forecast the Most Used Program Code for Each Month Next Year

We need a model that predicts the most frequent `ProgramCode` per month.

Here are **three approaches**:

- Model the monthly frequency of the dominant `ProgramCode` as a **time series**.
- Use **ARIMA** (AutoRegressive Integrated Moving Average) or **SARIMA** (Seasonal ARIMA) to capture trends and seasonality.
- Forecast next year's most frequent `ProgramCode`.

Pros:

- Captures temporal trends.
- Handles seasonality and long-term trends well.

Cons:

- Requires a sufficiently long historical dataset.
- Can struggle with sudden structural changes.



Approach 2: Machine Learning Classification (Random Forest/XGBoost)

- Treat this as a **classification problem** where the target is the most frequent `ProgramCode` per month.
- Use features like:
 - Month, Year, Economic Activity
 - Past frequencies of `ProgramCode`
 - Funding & Investment values
- Train a model (e.g., **Random Forest** or **XGBoost**) on past years and predict next year.



Pros:

- Handles categorical data well.
- Can incorporate multiple explanatory variables.

Cons:

- Requires extensive feature engineering.
- Less interpretable than time-series models.

Approach 3: Deep Learning (LSTM/RNNs)

- Use a **Recurrent Neural Network (RNN)** or **LSTM (Long Short-Term Memory)** to predict the dominant program.



Pros:

- Effective for long-term sequential data.
- Learns complex patterns.

Cons:

- Requires a large dataset.
- Computationally expensive.

Best Approach: Time Series Forecasting (SARIMA)



Given the nature of the problem—predicting **monthly trends**—SARIMA is the best choice. It efficiently captures trends and seasonality in categorical time-series data, making it more interpretable and accurate for forecasting.

Next Steps

1. Identify the state with the highest variation in `ProgramCode` frequency.
2. Build a SARIMA model to forecast the most used `ProgramCode` for each month next year.

I'll start by finding the state with the highest variation.

The state with the **highest variation** in its most frequent program code is **StateCode 25**.

Now, I'll analyze the monthly trend of its most used `ProgramCode` and build a SARIMA model to forecast the most frequent program for each month next year. 



It will be **ProgramCode 2** consistently across all 12 months.

Here's the Python code used for this analysis:

```
python
from statsmodels.tsa.statespace.sarimax import SARIMAX

# Filter data for the selected state (StateCode 25)
state_data = most_frequent_per_state_month[most_frequent_per_state_month["StateCode"] == state_wi

# Aggregate to get the most frequent ProgramCode per month
monthly_program_trend = state_data.groupby("MonthIssued")["ProgramCode"].agg(lambda x: x.value_co

# Prepare data for time series modeling (monthly frequency)
ts_data = monthly_program_trend.resample("M").first()

# Fit a SARIMA model to forecast next year's most used ProgramCode
sarima_model = SARIMAX(ts_data, order=(1,1,1), seasonal_order=(1,1,1,12), enforce_stationarity=False)
sarima_fit = sarima_model.fit()

# Forecast the most frequent ProgramCode for each month next year (assuming 12 months ahead)
forecasted_programs = sarima_fit.predict(start=13, end=24).round().astype(int)
print(forecasted_programs)
```

This result suggests **ProgramCode 2** has strong historical consistency in its frequency, leading to a stable forecast. If needed, we can refine this with additional factors like economic conditions or funding changes. Let me know if you want further analysis! [\[>-\]](#)

Message Data Analyst



ChatGPT can make mistakes. Check important info.